SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	CodePulse
Selected Domain	Domain 1
Selected SDG Track (1–17)	4,5,8,10
Project / Solution Name	Matchykan FutureReady
Name (Leader)	Anis Sofia
Email & Phone (Leader)	anissofiamohamadsalim@gmail.com +6018-774 6081
University Affiliation(s)	University Technology Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is transforming the skills required across industries, creating challenges for workers, fresh graduates, career switchers, and human resource (HR) professionals responsible for workforce development. As job requirements evolve, individuals may struggle to identify which skills remain relevant, which emerging capabilities to develop, and which career pathways offer suitable opportunities. Traditional career guidance can be fragmented across job platforms, training providers, and online resources, making it difficult to connect current capabilities with future skill requirements.

The problem also affects HR and learning-and-development managers, who need to identify workforce skill gaps and plan effective reskilling initiatives. Without accessible, evidence-informed tools, career planning can become time-consuming, while training decisions may not adequately reflect changing industry signals.

MatchyKan FutureReady addresses this challenge by connecting skills assessment, career exploration, AI-related trend signals, and reskilling guidance within one web application. Rather than treating career planning as a one-time job search, the proposed solution helps users understand their existing strengths, explore potentially suitable future roles, and identify skills they may need to develop.

The project supports Sustainable Development Goal (SDG) 4 through lifelong learning and relevant skills development, and SDG 8 through its focus on employability and productive work. Its objective is to make career development more accessible, understandable, and actionable while keeping people in control of their career decisions. It provides decision support rather than guaranteeing employment outcomes or claiming to predict the labour market with certainty.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

MatchyKan FutureReady is an AI-assisted career intelligence web application designed to connect an individual's current skills with potential future career opportunities and targeted reskilling actions. It

combines skills profile intake, career matching, AI-related trend signals, and career guidance in a unified interface.

Users can enter their skills and interests or provide a sample CV. With appropriate consent, the backend extracts relevant text and uses a RunPod Serverless endpoint running Qwen3-8B to suggest skills. Users can review these suggestions before adding them to their profile. The Career Matches component helps users explore potentially suitable roles through estimated skill overlap and identifies areas for further development.

The Skills Radar component uses public RSS news-feed samples and saved trend snapshots to present keyword-based signals about emerging AI and workforce topics. Source attribution and dates are important to distinguish observable evidence from assumptions. These signals should inform, rather than independently determine, career recommendations.

The proposed Phase 2 enhancement will connect trend signals to two explicitly labelled future scenarios: moderate and rapid AI disruption. Each scenario will use stated assumptions to explore how role priorities and skill-development needs might change. The reskilling pathway will then translate identified gaps into ordered learning steps, estimated completion times, and verified free or low-cost resources. Career recommendations and relevant skills will also be mapped to appropriate UN SDG targets with explanations of their potential contributions.

The current technical foundation uses React, TypeScript and Vite for the frontend, with Node.js and Express for backend services. RunPod provides AI inference, while public RSS feeds supply trend-related information. The team intends to apply SDC workshop learning in generative AI integration, database and data architecture, secure API handling, and cloud deployment.

MatchyKan FutureReady is intended to support informed human decisions, not replace professional judgement. Phase 2 validation will focus on recommendation traceability, scenario behaviour, usability, response reliability, and responsible handling of synthetic demonstration data.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: React, TypeScript and Vite for the web interface, including career exploration, skills visualisation and bilingual user guidance.

Backend: Node.js and Express with TypeScript for API requests, CV text extraction, AI inference requests and server-side handling of credentials.

AI model and inference: Qwen3-8B through RunPod Serverless for AI-assisted skill suggestions and career-coach responses. The API key must remain in server-side environment variables or secrets.

Trend data: Public RSS news-feed samples and saved snapshots provide keyword-based trend signals. Source links, publication dates and retrieval timestamps should be retained wherever available. These signals are not equivalent to verified job-vacancy statistics.

Profile storage: The supplied prototype uses browser-side storage for profile or progress information. A dedicated production database and associated access controls will need to be established if persistent multi-user storage is required.

Development and deployment: The application is developed within the existing project environment. A stable deployment URL, monitoring and production configuration should be confirmed before final demonstration.

System flow: User input → React frontend → Express backend → AI inference through RunPod and trend-data processing → career recommendations and skills-gap information → proposed reskilling pathway and SDG explanations.

Sponsor platform declaration: RunPod is used for model inference. Its status as an official SDC sponsor-provided platform should be verified against the event's official sponsor information .

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

MatchyKan FutureReady follows a user-centred design approach by focusing on the decisions that workers, graduates and career switchers need to make: understanding their current skills, identifying suitable roles, and deciding what to learn next.

The prototype brings profile intake, career matching, trend visualisation and AI-assisted guidance into one interface. Its design prioritises accessible navigation, understandable recommendations, and user control over suggested skills. CV submission is optional, and demonstration activities will use synthetic profiles rather than real personal CVs, in accordance with the Domain 1 requirements.

Technical feasibility will be evaluated through end-to-end testing of the profile-to-match journey, AI endpoint reliability, trend-source availability, response latency and recommendation traceability. A key technical risk is that keyword-based news signals may not accurately represent actual hiring demand. The mitigation strategy is to display source evidence and dates, distinguish observed signals from forecasts, and provide a fallback when data sources fail.

For Phase 2, the team will conduct usability testing with consenting participants using synthetic demonstration data, establish baseline task-completion measurements, and refine the interface based on feedback. Deployment readiness will require secure server-side credentials, documented setup, a stable hosted application, clear privacy messaging, and a GitHub repository that matches the demonstrated version. Future scenario modelling and personalised learning recommendations will be validated before being presented as reliable decision-support features.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary target: SDG 8.5** — Achieve full and productive employment and decent work for all, including young people. MatchyKan FutureReady supports career exploration and skills-development planning that may help users prepare for changing workforce requirements. **Supporting target: SDG 4.4** — Increase the number of youth and adults with relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship. The proposed personalised reskilling pathway aims to help users identify skill gaps and access relevant learning resources. The project's contribution is indirect and will be evaluated through usability and skills-planning outcomes, rather than claims of directly creating jobs or achieving SDG targets.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	A verified numerical baseline for MatchyKan's target users has not yet been established. The project will therefore avoid presenting an unsupported national skills-gap percentage or claiming that the prototype has already improved career outcomes. During Phase 2, the team proposes a pilot involving at least 10 consenting student or early-career participants, subject to recruitment feasibility. Participants will complete a standardised career-exploration task using

Metric Category	Your Response
	synthetic demonstration profiles. The team will record baseline median task-completion time, task-completion rate, and participants' ability to explain why a role was recommended. Results will be reported with the sample size and study method and will not be generalised to the entire Malaysian workforce.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	**Primary KPI:** Median time, in minutes, required to complete a defined career-exploration task, from entering skills to identifying a suitable role and a relevant next learning step. **Secondary KPIs:** ● Task-completion rate (%). ● Percentage of participants who can correctly explain the basis of a role recommendation. ● Percentage of recommendations with accessible supporting sources and dates. Measurements will be recorded during the Phase 2 pilot. These indicators assess usability and transparency, not actual job placement or long-term employment outcomes.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	The proposed Phase 2 target is to reduce median task-completion time by 20% relative to the baseline pilot, while maintaining or improving task-completion rate. The team will use the same task definition and comparable testing conditions for baseline and follow-up measurements. Actual results, sample size and limitations will be reported at the final presentation. The 20% figure is a prospective project target, not an achieved result or an externally validated impact estimate.

5. Eligibility & Readiness Checklist

Tick all

boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.

- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ✓ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private ✓ datasets, or pre-trained AI models that your team did not create?

Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React, React DOM, Vite and TypeScript	Software libraries	Verify individuals package and licenses
Node.js and Express	Runtime and framework	Verify applicable software licenses
RunPod serverless	Hosted inference service	Applicable service terms
Qwen3-8B	Pre trained AI models	Verify the exact model card and licenses
Public RSS sources	Third party content	Applicable source site terms and attribution requirements

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ✓ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

24.10.2026

[Date]

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